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MATHEMATICS

STANDARD SIX

Pre-annual joint assessment - November 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	04
Level	Standard Six
Year	2024
Exam Board / Issuer	Top Abilities
Exam	Pre-annual joint assessment - November
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts (8 main questions)

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-Top-Abilities-November-Annual-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Section A: Mathematical operations

1(i). Calculate $10199 + 11$.

Step 1: 1 place: $9 + 1 = 10$; write 0, carry 1.

Step 2: 10 place: $9 + 1 + \text{carry } 1 = 11$; write 1, carry 1.

Step 3: 100 place: $1 + 0 + \text{carry } 1 = 2$; write 2.

Step 4: 1000 place: $0 + 0 = 0$; write 0.

Step 5: 10000 place: $1 + 0 = 1$; write 1.

Step 6: $10199 + 11 = 10210$.

Final Answer: 10210

1(ii). Calculate 791×88 .

Step 1: Split into place values: $700 \times 88 = 61600$; $90 \times 88 = 7920$; $1 \times 88 = 88$.

Step 2: Add the partial products: 69608.

Final Answer: 69608

1(iii). Calculate $90000 - 3856$.

Step 1: 1 place: regroup 1 from the next place; $10 - 6 = 4$.

Step 2: 10 place: regroup 1 from the next place; $9 - 5 = 4$.

Step 3: 100 place: regroup 1 from the next place; $9 - 8 = 1$.

Step 4: 1000 place: regroup 1 from the next place; $9 - 3 = 6$.

Step 5: 10000 place: $8 - 0 = 8$.

Step 6: $90000 - 3856 = 86144$; check $86144 + 3856 = 90000$.

Final Answer: 86144

1(iv). Calculate $18415 \div 29$.

Step 1: $184 \div 29$: write 6; $6 \times 29 = 174$; $184 - 174 = 10$.

Step 2: $101 \div 29$: write 3; $3 \times 29 = 87$; $101 - 87 = 14$.

Step 3: $145 \div 29$: write 5; $5 \times 29 = 145$; $145 - 145 = 0$.

Step 4: $29 \times 635 = 18415$.

Step 5: $18415 \div 29 = 635$.

Final Answer: 635

1(v). Calculate $8 \frac{2}{3} + 7 \frac{1}{4}$.

Step 1: Convert the mixed numbers: $(8 \times 3 + 2)/3 = 26/3$; $(7 \times 4 + 1)/4 = 29/4$.

Step 2: Use denominator 12: $26/3 = 104/12$ and $29/4 = 87/12$.

Step 3: $104/12 + 87/12 = 191/12 = 15 \frac{11}{12}$.

Final Answer: $15 \frac{11}{12}$

1(vi). Calculate 0.729×7.04 .

Step 1: First calculate whole-number products: $729 \times 700 = 510300$ and $729 \times 4 = 2916$.

Step 2: $510300 + 2916 = 513216$.

Step 3: The factors have $3 + 2 = 5$ decimal places in total, so the product is 5.13216.

Final Answer: 5.13216

1(vii). Calculate $1 - 0.3094$.

Step 1: Write 1 as 1.0000 and align decimal places.

Step 2: Regroup through the zeros to give 9 tenths, 9 hundredths, 9 thousandths and 10 ten-thousandths.

Step 3: Subtract from the right: $10 - 4 = 6$; $9 - 9 = 0$; $9 - 0 = 9$; $9 - 3 = 6$.

Step 4: Therefore $1.0000 - 0.3094 = 0.6906$.

Final Answer: 0.6906

1(viii). Calculate $-107 - 68 + 2$.

Step 1: $-107 - 68 = -175$.

Step 2: $-175 + 2 = -173$.

Final Answer: -173

1(ix). Calculate $0.258 \div 0.05$.

Step 1: Multiply both numbers by 1000 to remove decimal points: $0.258/0.05 = 258/50$.

Step 2: $50 \times 5 = 250$, leaving $258 - 250 = 8$.

Step 3: $8/50 = 0.16$, so $258/50 = 5.16$.

Final Answer: 5.16

1(x). Calculate the printed mixed-number forms $3 \frac{5}{3} \div 9 \frac{9}{4}$.

Step 1: Read the printed mixed forms literally: $3 + \frac{5}{3} = \frac{14}{3}$ and $9 + \frac{9}{4} = \frac{45}{4}$.

Step 2: Divide by multiplying by the reciprocal: $(\frac{14}{3}) \times (\frac{4}{45})$.

Step 3: $14 \times 4 = 56$ and $3 \times 45 = 135$; the fraction $\frac{56}{135}$ is already simplest.

Final Answer: $\frac{56}{135}$

Section Summary: Align place values and decimal points. Convert mixed numbers to improper fractions; use a common denominator for addition and subtraction and the reciprocal for division.

Section B: Word problems

2(i). Form the smallest four-digit numeral divisible by 5 from the digits 2, 7, 0, 5, 6.

Step 1: A number divisible by 5 must end in 0 or 5.

Step 2: Using each selected digit once, the smallest numeral ending in 5 is 2065; the smallest ending in 0 is 2560.

Step 3: $2065 < 2560$, so the intended distinct-digit answer is 2065.

Step 4: The question does not explicitly prohibit repetition. If repetition is allowed, 2000 is the smallest possible numeral.

Final Answer: 2065 using distinct digits; 2000 if repeated digits are allowed.

2(ii). A candidate received MCV votes from the town and XXXIV votes from the village. Find the total in Roman numerals.

Step 1: $MCV = 1000 + 100 + 5 = 1105$; $XXXIV = 30 + 4 = 34$.

Step 2: $1105 + 34 = 1139$.

Step 3: $1139 = 1000 + 100 + 30 + 9 = M + C + XXX + IX = MCXXXIX$.

Final Answer: MCXXXIX (1139 votes)

2(iii). Fill the missing number: 34, 68, 102, __, 170.

Step 1: $68 - 34 = 34$ and $102 - 68 = 34$, so add 34 each time.

Step 2: $102 + 34 = 136$; check $136 + 34 = 170$.

Final Answer: 136

2(iv). 6840 posters were shared equally among 20 towns. How many posters did 10 towns receive altogether?

Step 1: Each town received $6840/20 = 342$ posters.

Step 2: Ten towns received $342 \times 10 = 3420$ posters.

Final Answer: 3420 posters

2(v). Round 0.6379 to two decimal places.

Step 1: The hundredths digit is 3 and the next digit is 7.

Step 2: Since 7 is at least 5, raise the hundredths digit from 3 to 4.

Final Answer: 0.64

2(vi). Find the total value of the digit 5 in the product of 9 and 435.

Step 1: $9 \times 435 = 9 \times (400 + 30 + 5) = 3600 + 270 + 45 = 3915$.

Step 2: The digit 5 in 3915 is in the ones place, so its value is 5.

Final Answer: 5

3(i). Find the sum of the LCM and GCF of 12, 36 and 15.

Step 1: $12 = 2^2 \times 3$; $36 = 2^2 \times 3^2$; $15 = 3 \times 5$.

Step 2: $\text{LCM} = 2^2 \times 3^2 \times 5 = 180$; the only common prime factor is 3, so $\text{GCF} = 3$.

Step 3: $180 + 3 = 183$.

Final Answer: 183

3(ii). The three sisters are 19, 23 and 31 years old. Find the average age of only the two youngest sisters.

Step 1: The two youngest ages are 19 and 23 years.

Step 2: $19 + 23 = 42$; $42/2 = 21$.

Final Answer: 21 years

3(iii). A worker earns Sh 720000 and spends $\frac{1}{4}$ of his salary on rent. The remainder is shared equally for the school fees of his 12 children. Find the fee for one child.

Step 1: $\text{Rent} = 720000 \times \frac{1}{4} = \text{Sh } 180000$.

Step 2: $\text{Remaining salary} = 720000 - 180000 = \text{Sh } 540000$.

Step 3: $\text{One fee} = 540000/12 = \text{Sh } 45000$.

Final Answer: Sh 45000

4(i). Joharia deposited Sh 600000 at 6% interest per year. After how many years will she earn Sh 90000 interest?

Step 1: Using simple interest, $\text{interest for one year} = 600000 \times \frac{6}{100} = \text{Sh } 36000$.

Step 2: $\text{Time} = 90000/36000 = \frac{5}{2} = 2.5 \text{ years}$.

Step 3: Half a year is 6 months.

Final Answer: 2.5 years (2 years 6 months)

4(ii). A plane travels at 7324.73 km/h for 7 hours. Find the distance travelled.

Step 1: $\text{Distance} = \text{speed} \times \text{time}$.

Step 2: $7324.73 \times 7 = 51273.11$.

Final Answer: 51273.11 km

4(iii). How many 0.975 are there in 156?

Step 1: Number of groups = $156/0.975$.

Step 2: Multiply numerator and denominator by 1000: $156000/975 = 160$.

Step 3: Check $0.975 \times 160 = 156$.

Final Answer: 160

5(i). An engineer can reach a ceiling bulb 9 m above the ground while standing on a table 750 cm high. Find his height.

Step 1: Convert the table height: $750/100 = 7.5$ m.

Step 2: Under the intended table-plus-person model, height = $9 - 7.5 = 1.5$ m.

Step 3: Reaching height and body height differ in practice; the printed information only directly determines a 1.5 m reach above the table.

Final Answer: 1.5 m under the intended elementary model; body height is not independently determined.

5(ii). Find k if $7(k - 1) = -k + 9$.

Step 1: Expand: $7k - 7 = -k + 9$.

Step 2: Add k to both sides and add 7 to both sides: $8k = 16$.

Step 3: $k = 16/8 = 2$.

Step 4: Check: $7 \times (2 - 1) = 7$ and $-2 + 9 = 7$.

Final Answer: $k = 2$

5(iii). A number multiplied by 8 and then increased by 15 equals 22. Find the number.

Step 1: Let the number be n: $8n + 15 = 22$.

Step 2: $8n = 22 - 15 = 7$.

Step 3: $n = 7/8$.

Step 4: Check: $8 \times (7/8) + 15 = 22$.

Final Answer: $7/8$

6(i). Evaluate x^2y when $x = 4/5$ and $y = 2/3$.

Step 1: $(4/5)^2 = 16/25$.

Step 2: $(16/25) \times (2/3) = (16 \times 2)/(25 \times 3) = 32/75$.

Final Answer: 32/75

6(ii). A circle has circumference 423.9 m. Find its diameter using $\pi = 3.14$.

Step 1: Circumference = $\pi \times \text{diameter}$, so diameter = $423.9/3.14$.

Step 2: Since $3.14 \times 135 = 423.9$, the diameter is 135 m.

Final Answer: 135 m

6(iii). A parallelogram has base 36 cm and perpendicular height 12 cm. Find its area.

Step 1: Area = base $\times$ perpendicular height.

Step 2: $36 \times 12 = 432$.

Final Answer: 432 cm²

Section Summary: Choose the operation from the question, convert to consistent units, and check the result. LCM uses the highest prime powers; GCF uses only factors shared by all numbers.

Section C: Geometry and statistics

7(i). 14400 animals were illegally hunted in Manyara National Park between January and October 2024. The pie chart labels buffaloes 90° (right-angle mark), zebras 100° and lions 50° . How many antelopes were hunted?

Step 1: A full pie chart is 360° .

Step 2: Antelope angle = $360 - 90 - 100 - 50 = 120^\circ$.

Step 3: Antelope fraction = $120/360 = 1/3$.

Step 4: $14400 \times 1/3 = 4800$.

Final Answer: 4800 antelopes

7(ii). Two cylindrical concrete tanks each have radius 7 m and height 50 m. Find their combined capacity in cubic metres.

Step 1: Volume of one tank = $\pi \times \text{radius}^2 \times \text{height} = \pi \times 7^2 \times 50 = 2450 \pi \text{ m}^3$.

Step 2: Combined capacity = $2 \times 2450 \pi = 4900 \pi \text{ m}^3$.

Step 3: If $\pi = 22/7$ is used, capacity = $4900 \times 22/7 = 15400 \text{ m}^3$.

Final Answer: $4900 \pi \text{ m}^3$ (15400 m^3 using $\pi = 22/7$)

8(i). A semicircle has diameter 28 cm and its right half is shaded. Find the shaded area.

Step 1: Radius = $28/2 = 14 \text{ cm}$.

Step 2: The shaded part is one quarter of a full circle.

Step 3: Area = $\pi \times 14^2/4 = 49 \pi \text{ cm}^2$.

Step 4: If $\pi = 22/7$ is used, $49 \times 22/7 = 154 \text{ cm}^2$.

Final Answer: $49 \pi \text{ cm}^2$ (154 cm^2 using $\pi = 22/7$)

8(ii). A right triangle has hypotenuse 10 cm and vertical leg 6 cm. Find its area.

Step 1: $\text{Base}^2 = 10^2 - 6^2 = 100 - 36 = 64$.

Step 2: The positive base length is 8 cm because $8^2 = 64$.

Step 3: Area = $8 \times 6/2 = 24$.

Final Answer: 24 cm^2

8(iii). Write the coordinates of points A and B on the grid.

Step 1: Read the horizontal coordinate first and the vertical coordinate second.

Step 2: A is 1 unit left and 3 units above the origin, so $A = (-1, 3)$.

Step 3: B is 2 units right and 2 units below the origin, so $B = (2, -2)$.

Final Answer: $A = (-1, 3)$; $B = (2, -2)$.

Section Summary: Use marked measurements and angles. Circle sectors are fractions of 360° . Right triangles obey $a^2 + b^2 = c^2$; cylinder volume is $\pi \times r^2 \times h$. Read graph coordinates as (x, y) .

Source Checks

Darasa Huru - TOP ABILITIES STANDARD SIX PRE-ANNUAL JOINT EXAMS 2024